



Intact or impaired? The understanding of give-and-take interactions in children with autism spectrum disorder[☆]

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ABSTRACT

Background: Understanding and predicting others' behavior in a dynamic and rapidly changing world is a fundamental aspect of social interactions. However, it remains unclear as to whether children with autism spectrum disorder (ASD) could understand and predict goal-directed social actions.

Aims: To investigate the understanding of give-and-take interactions in children with ASD with the use of eye tracking.

Methods and procedures: Experiment 1 and 2 investigated the understanding of giving and taking respectively in 5- to 8-year-old Chinese children with ASD and typically developing children by using the eye-tracking technology.

Outcomes and Results: We found that children with ASD could predict actions, but they were less proficient in processing give-and-take interactions. Moreover, children with ASD showed impaired understanding of giving but not taking.

Conclusions: These results indicate that the basic mechanisms of action prediction are intact in children with ASD whereas there may be deficits in the top-down social processing of the giving gesture.

What this paper adds

Understanding the actions and intentions of others is essential for social interactions. The current study was the first to investigate autistic children's understanding of goal-directed social actions by recording their eye movements in perceiving give-and-take interactions. We found that children with ASD show intact understanding of taking but not giving, reflecting a dissociation of understanding of giving and taking. Further, we found that atypical understanding of giving is negatively correlated with social communication ability in children with ASD. These findings shed light on the understanding of social actions in children with ASD.

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1. Introduction

Understanding and predicting others' goal-directed action in a dynamic and rapidly changing world is crucial for success in social interactions (Gredebäck & Falck-Ytter, 2015; Hudson et al., 2012). From infancy, typically developing (TD) children show sensitivity to social cues such as gaze direction, gestures and are able to use these social cues to understand others' intentions and anticipate their behaviors in the immediate future (Falck-Ytter, Gredebäck, & von Hofsten, 2006; Flanagan & Johansson, 2003). Understanding goal-directed actions enables infants to learn about actions and objects through observation and interaction, and to engage in relatively complex non-verbal communicative interactions (Gredebäck & Melinder, 2010a; Thorgrímsson, Fawcett, & Liszkowski, 2014).

Autism Spectrum Disorder is a developmental disability with impaired social interaction and communication as its core feature (American Psychiatric Association, 2013). It has been proposed that social impairments in individuals with ASD may result from impaired action anticipation (Sinha et al., 2014). However, the evidence is mixed as to whether children with ASD could understand and predict others' behavior. Some studies suggest that children with ASD can understand goal-directed actions. For instance, Aldridge et al. (2000) revealed that children with ASD aged between 2 years 2 months and 4 years 2 months could complete the unfulfilled actions performed by the experimenters by using a behavioral re-enactment paradigm in which an adult demonstrated an intention to manipulate the object in a certain way but failed, then the experimenter presented the child the same object to examine whether he/she could complete the unfulfilled actions performed by the adult. With the use of eye tracking technology, several studies found that 5-year-old (Falck-Ytter, 2010) and 10-year-old (Ganglmayer et al., 2020) children with ASD could anticipate goal-directed actions. In a gesture recognition task, 8-year-old children with ASD showed superior performance compared with TD children of the same verbal mental age, suggesting that action understanding in children with ASD might be intact (Hamilton et al., 2007).

On the contrary, accumulating evidence reveals that children with ASD show impairments in understanding and predicting goal-directed actions (Krogh-Jespersen et al., 2018; Somogyi et al., 2013). For example, Krogh-Jespersen et al. (2018) investigated the action prediction in 2-year-old children with ASD, a chronological age-matched group of TD children and a mental age-matched group of TD children (17-month-olds). They presented children with a video of goal-directed actions in which an actress reached and grasped one of two objects in the familiarization stage. In the test stage, the actress swapped the location of the target object and the non-target object and did not provide trajectory information. It was found that children with ASD looked at the prior location of the target object instead of the target itself while TD children looked at the target object in the test stage, suggesting that children with ASD predict others' action based on kinematic information instead of the action goal. In addition, some researchers found that 10-year-old children with ASD have difficulty using frequency information to generate action predictions (Schuwerk et al., 2016). However, there is evidence suggesting that the ability to understand others' actions is not entirely impaired in children with ASD. Vivanti et al. (2011) revealed that 13-year-old children with ASD could predict others' actions based on the facial emotional expression and rationality of actions whereas they failed to use cues like direct gaze and head-turn to predict others' action. It indicates that children with ASD have difficulty interpreting the meaning of referential cues and using these cues to anticipate other people's actions. Due to the different results regarding the action understanding in children with ASD, it remains unclear whether children with ASD predict others' actions based on the action goal or they adopt a compensatory strategy, such as reliance on trajectory information or the function of objects to predict actions (Ganglmayer et al., 2020).

In addition, aforementioned studies investigating action prediction in children with ASD primarily focused on non-social actions, yet less is known about their ability to understand and predict social actions that are embedded with communicative meaning. Only a few studies explored the understanding of social actions in children with ASD. For example, Centelles et al. (2013) presented 6–12-year-old children with ASD and 4–10-year-old TD children point-light displays in which two actors were either interacting with each other (social condition) or moving side by side without interacting (non-social condition) and all children were asked to judge whether the two actors were communicating or not in both conditions by pressing the colored button on the key board. Results showed that children with ASD were less accurate than TD children in both conditions but their correct response rate was above the chance level. It is unknown, however, whether children with ASD's decision was based on the action goal or the proximity of the two actors. In the social interaction condition, the proximity between the actors was closer than that in the non-social interaction condition. Thus, the evidence remains unclear because of confounding variables in previous research such as a lack of salient differences between social and non-social conditions.

In the current study, we investigated children with ASD's understanding of goal-directed social actions by recording their eye movements in perceiving give-and-take interactions as eye tracking technology allows researchers to measure online action understanding of children through their eye movements (Krogh-Jespersen et al., 2018). Giving and taking are two important social actions in social communication (Hay & Murray, 1982; Mundy et al., 1986), and linked to the development of social cognitive skills such as discrimination between self and others and joint attention (Mundy et al., 1986). The giving action represents a spontaneous transfer of ownership and is also the basis for higher-level behaviors such as resource exchange and distribution, gifting, and sharing, and plays an important role in the social development of early childhood (Juvrud et al., 2018). The gesture of taking holds many communication functions, such as directing an object, expressing a request, and conveying an invitation to cooperate (Elsner et al., 2014; Mundy et al., 1986). In the first year, infants are sensitive to the social meaning of giving and taking. Twelve-month-old infants possess an action schema of giving and represent giving as an object-mediated social interaction (Tatone et al., 2015). Typically developing babies are sensitive to the taking gesture and show increased P400 amplitude, an ERP component sensitive to the congruency of gestures such as pointing (Gredebäck and Melinder, 2010b) or grasping (Bakker et al., 2015), when they are presented with the taking gesture compared with a non-communicative hand shape at 9 months of age (Bakker, Kaduk et al., 2015). When observing the give-and-take interaction, 12-month-old infants show faster predictive saccades to the taking gesture than an inverted hand shape without communicative meaning (Elsner et al., 2014). At the age of 2–3, children can spontaneously use the give-and-take action sequence to

share objects (Juvrud et al., 2018). There is evidence that children with ASD show lower level of sharing (Dunfield et al., 2019), and it is unknown whether reduced levels of sharing is associated with understanding of give-and-take interactions, especially giving.

The investigation of giving and taking in children with ASD can provide valuable insights. For researchers, knowing whether children with ASD can understand giving and taking might contribute to elucidating the underlying mechanisms of social difficulty (i. e., impaired cooperation, sharing) in children with ASD. Clinicians can benefit from this information in assessing ASD children's comprehension of social gestures and developing new treatment goals to facilitate social interactions of children with ASD. However, to our knowledge, no studies have explored the understanding of giving in children with ASD. And only one study indirectly examined 4–7-year-old autistic children's understanding of taking gesture and found that children with ASD showed less sensitivity to the taking gesture (Falck-Ytter et al., 2013). This study (Falck-Ytter et al., 2013) investigated autistic children's visual attention patterns by presenting them with videos in which a child requested an older child's toy using a taking gesture but he was ignored. These researchers found that TD children tended to look at the older child's face after watching the requesting gesture while this tendency was weaker in children with ASD. However, from this research, it is difficult to identify whether children with ASD could understand taking because it is possible that ASD children's decreased attention to the older child's face is due to their diminished attention to social stimuli as evidenced by numerous studies (Chawarska et al., 2013; Moore et al., 2018; Wang et al., 2018). Therefore, further work is necessary to elucidate whether children with ASD could understand giving and taking.

Action prediction is a widely adopted measure to represent action understanding (Gredebäck & Falck-Ytter, 2015). One important question concerning action prediction in children with ASD is: do they learn the specific action goal or do they anticipate the spatiotemporal movement trajectories when they perceiving goal-directed actions (Ganglmayer et al., 2020)? However, merely presenting children with daily goal-directed actions cannot distinguish children's action prediction is based on the action goal or based on the trajectory movement of the hands. To address this question, in the present study, we presented children with ASD and TD children with rational giving and irrational giving (the configuration of the hand and arm is distorted and not embedded with social meaning) in experiment 1 to investigate children's understanding of giving. We hypothesized that children with ASD would show anticipatory looking in the rational giving condition but not in the irrational giving condition if they can understand giving. In experiment 2, we manipulated taking in the same way to examine children's understanding of taking. We hypothesized that if they could understand taking, children with ASD would exhibit faster saccades in the rational taking condition than in the irrational taking condition. Moreover, we aimed to explore the relationships between understanding of giving and taking and the autistic behaviors in children with ASD.

2. Experiment 1

2.1. Method

2.1.1. Participants

Forty-three Chinese children with ASD and forty TD children participated in the giving task. Among them, 4 TD children and 7 ASD children were excluded due to insufficient valid gaze shifts over all trials because of inability to keep attentive through the experiment and calibration failure. The final sample consisted of 36 children with ASD (mean age = 5.99 years) and 36 TD children between 5 and 8 years old (mean age = 6.13 years). The TD and ASD children did not differ in terms of age and gender composition (see Table 1). All children with ASD were diagnosed by professional psychiatrists by using DSM-5 and met the diagnostic criteria. Children with ASD were recruited from Peking University Sixth Hospital and several special education centers for children with ASD in multiple places across China (Beijing, Qingdao, Guangzhou and Shenzhen) to ensure the sample is representative. TD children were recruited via kindergartens and primary schools and had no history of developmental disorders. Informed consent was obtained from parents and oral consent was received from children. This study has been approved by the local Child Subjects Ethics Committee and conformed to the ethical standards set by the Declaration of Helsinki.

Table 1
Demographic characteristics of participant groups in experiment 1.

	ASD (<i>n</i> = 36) M (SD)	TD (<i>n</i> = 36) M (SD)	<i>p</i> value
Sex, M/F	30/6	29/7	0.763
Age (years)	5.99 (0.88) Range: 5–8.07	6.13 (0.92) Range: 5.01–8.03	0.504
IQ	112 (14.88)	122 (14.52)	0.01 ^a
AQ-Child	Range: 90–138	Range: 95–142	
SRS	86.41 (14.54) Range: 60–113	-	
	86.52 (24.96) Range: 31–138	-	

Note. AQ: Autism Spectrum Quotient-Children's Version, SRS: Social Responsiveness Scale

^a $p \leq .01$.

2.1.2. Assessments

Nonverbal intelligence was measured with the Standard Raven Test. To further confirm autism severity levels and social skills of children with ASD, caregivers of ASD children filled out the Autism Spectrum Quotient-Children's Version (AQ-Child, Auyeung et al., 2008) and Social Responsiveness Scale (SRS, Cen et al., 2017). The AQ-Child is a 50-item parent-report questionnaire, which consists of autistic behaviors across five domains: social skills, attention switching, attention to detail, communication, and imagination. Each domain includes ten items. The AQ score ranges from 0 to 150 with higher AQ score reflecting more autistic behaviors and the cut-off is 76 (Auyeung et al., 2008). The AQ-Child showed adequate internal consistency (Cronbach's $\alpha = 0.786\text{--}0.840$) and test-retest reliability (0.536–0.791) in the mainland China (Sun et al., 2019). The SRS is a 65-item questionnaire including five subscales: social awareness (8 items), social cognition (12 items), social communication (22 items), social motivation (11 items) and autistic mannerisms (12 items). The SRS score ranges from 0 to 195 with higher SRS score indicating more severe social impairments. The total SRS raw scores can be transformed into T-scores and the cut-off is 60. In the current study, we used raw SRS scores to maximize phenotypic variability in children with ASD. The SRS also showed satisfactory internal consistency (Cronbach's $\alpha = 0.871\text{--}0.922$) and test-retest reliability (0.81–0.94) (Cen et al., 2017).

2.1.3. Stimuli and apparatus

Children's eye movements were recorded using the Tobii X3–120 eye tracker with a sampling rate of 120 Hz. Based on the experimental paradigm in the Elsner et al. (2014)'s study, we added a pointing action before the give-and-take interactions to make the scene with more ecological validity. In addition, we aimed to investigate whether adding a pointing stage could serve as a prompt for children to predict the upcoming give-and-take interaction since pointing is a step between reaching for an object and manipulating the social partner to obtain the object (Crais, Douglas, & Campbell, 2004). All children were presented with 12 videos in which two actresses sat at a table. On the side of the actress A (giver) was a toy. The actress B (taker) pointed to the toy, and then performed a taking gesture. The actress A grasped and gave the toy to actress B. In half the trials, the giver performed the giving action in a rational manner while in another half of trials, the giver performed the giving action in an irrational manner (see Fig. 1). The two actresses in the video did not talk in order to reduce the challenges of language processing in children with ASD. The position of the actress was counterbalanced and the videos were presented randomly to children. Each video lasted approximately 10s.

2.1.4. Procedure

Before the experiment, the experimenter told children that they would be presented with a series of videos on the computer screen and what they need to do is to watch these videos carefully. The experiment was conducted in a quiet room in the kindergarten, primary school, hospital or autism centers. Children were seated on a comfortable seat approximately 60 cm in front of the eye tracker. A five-point calibration procedure was performed to calibrate the eye-tracker, with one point in each of the four screen corners and one point in the middle of the screen. If four or more points were calibrated successfully, the child entered the experiment session. Otherwise, the calibration procedure would be repeated (Braukmann et al., 2018). After a standard 5-point calibration, children were

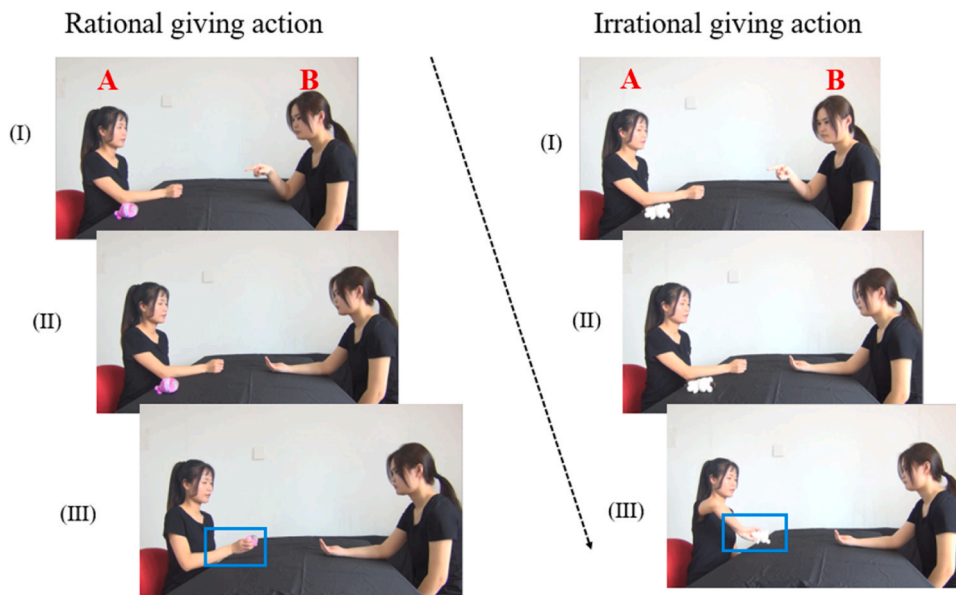


Fig. 1. Video clips of action sequences presented in experiment 1. Note. In both rational giving condition and the irrational giving condition, the actress B first pointed to the toy (I), then the actress B showed a taking gesture (II), the actress A gave the toy to the actress B (III). The difference between the two conditions is that in the rational giving condition, the actress A grasped the toy and gave the toy in a rational manner (the action is marked by the blue rectangle in the left side) while in the irrational giving condition, the actress A grasped the toy and passed the toy in an irrational manner (the configuration of the hand and arm is distorted, the action is marked by the blue rectangle in the right side).

presented with videos of give-and-take interaction, alternating giving from the right or the left between consecutive trials. Each child was then shown 12 trials with 6 rational giving and 6 irrational giving. After completing the eye tracking task, children were administered with the Standard Raven Test by the experimenter. Parents of children with ASD filled out the AQ and SRS questionnaire under the guidance of a trained experiment assistant. The whole procedure lasted about 1 h and children were given a gift at the end of the experiment.

2.1.5. Data processing

Given that the give-and-take interactions in our study consisted of two stages: pointing (pre-giving stage) and give-and take interaction, we analyzed data of the two stages separately. In the pointing stage, we aimed to investigate whether children have expectations for the upcoming social interaction by using the cues conveyed by the interactors before the formal interaction begins, and whether this expectation would influence children's action prediction in the next stage of interaction. The second stage was the give-and-take interaction, which was designed to examine whether children could understand giving. In the pointing stage, the main areas of interest (AOIs) were the face of the target of pointing (the face of the actress A) and the face of the performer of pointing (the face of the actress B), the action of pointing (hand) and the object (toy), see Fig. S1. In the pointing stage, two dependent measures were extracted from the gaze data. (1) The first fixation time. This index measures the time it takes for the child to first notice the AOIs after the time window of the area of interest begins, which intended to explore whether the child is sensitive to pointing. (2) The proportional total looking time. This indicator measures the overall gaze time of the children to each interest area after the time window of the interest area starts, and the purpose is to investigate whether the child would spend more time processing important cues. The proportional looking time measure was calculated by dividing the looking time to the target AOI by the total looking time to all AOIs. In the give-and-take stage, the AOIs were shown in Fig. S2, and one dependent variable was the proportional total looking time. Besides that, we used the difference score to measure the action understanding of children. Difference score was obtained by subtracting the time when children first gazed at the taking hand from the time when the hand of the giver touched the taker's hand. Positive numbers refer to children shifting their gaze to the taking hand before the giving hand arriving at the taking hand, which suggests that children show anticipatory looking and are able to predict other's actions, and negative numbers refer to children shifting their gaze to the taking hand after the arrival of the giving hand to the taking hand, which suggests that children do not exhibit predictive saccade and merely follow other's actions. Faster predictive saccade in the rational giving condition means a better understanding of giving. All included children exhibited gaze shifts in at least 6 of 12 presented trials.

2.2. Results

Can children with ASD follow the pointing gesture?

The first fixations to action of pointing (hand), the object (toy), and the target of pointing (The face of the actress A).

Since the ASD group and TD group significantly differed in IQ, the one-way ANCOVA with IQ as a covariate was performed to evaluate children's first looking time to the pointing scene. There were no significant differences in the first looking time to the toy ($p = .847$), the pointing action ($p = .824$) and the target of pointing (The face of the actress A) ($p = .077$) between TD children and children with ASD, indicating that both groups of children could follow the direction of pointing.

The proportional total looking time to pointing action (hand), the object (toy), the target of pointing (the face of the actress A), and the performer of pointing (the face of the actress B).

The results of the one-way ANCOVA with IQ as a covariate revealed that ASD group and TD group did not differ in terms of the proportional total looking time to the pointing action ($p = .147$), the target of pointing (the face of actress A) ($p = .3$), the performer of pointing (the face of actress B) ($p = .293$) and the object (toy) ($p = .516$).

Can children with ASD understand the meaning of giving?

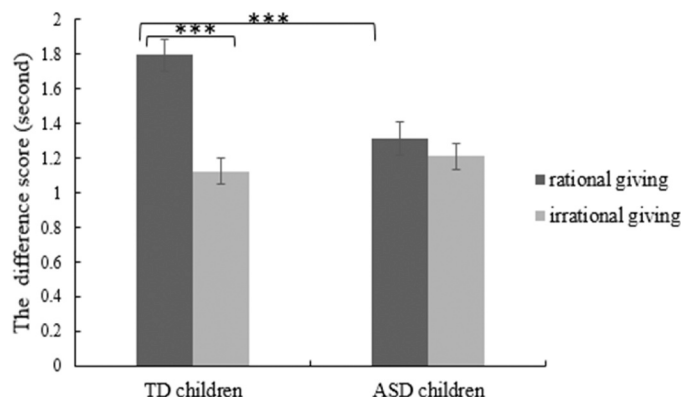


Fig. 2. Difference scores in the experiment 1. Note. The difference score was defined as the arrival time of the giving hand to the taking hand minus the arrival of gaze to the taking hand. Error bars indicate standard error. Asterisks represent statistically significant differences and interactions ($p < .001$).

Difference score of TD children and children with ASD.

To assess children’s performance of action prediction in the giving stage, the mixed repeated measures ANCOVA was conducted with group (TD vs. ASD) as the between-subjects measure, condition (rational giving vs. irrational giving) as the within-subjects measure and IQ scores as the covariate. The interaction between group and condition was significant ($F(1,69) = 9.318, p = .003, \eta^2_p = .119$). The difference scores of two groups of children were positive, indicating that both children with ASD and TD children show predictive eye movements. A simple effect analysis of the interaction found that in the condition of rational giving, TD children shifted their gaze to the taking hand faster than children with ASD, $F(1, 69) = 12.205, p = .001, \eta^2_p = .15$. In the condition of irrational giving action, the difference scores of two groups of children were not significantly different. In addition, TD children showed faster saccades in the rational giving condition than in the irrational giving condition, $F(1, 69) = 27.837, p < .001, \eta^2_p = .287$, while there was no significant difference in the difference scores of children with ASD in two conditions (see Fig. 2). In addition, overall, TD children showed faster predicative saccade than ASD children in both conditions ($F(1, 69) = 5.733, p = .019, \eta^2_p = .077$), indicating that children with ASD are less efficient in predicting actions. In terms of the main effect of condition, no significant differences were found.

Percentage of total looking time to the giving hand, the taking hand and the faces of two actresses in the rational giving and irrational giving condition.

The results of mixed repeated measures ANCOVA showed that there were no significant differences in the proportional total looking time to the giving hand, the taking hand and the faces of two actresses between two groups of children in the two conditions (all $ps > .05$).

Is children’s following of pointing correlated with their understanding of giving?

Partial correlation analysis with IQ as a control variable was conducted to investigate whether children’s following of pointing would influence their understanding of giving in the give-and-take interaction. We used the first fixation to the toy to measure children’s following of pointing. Shorter first fixation to the toy in the pointing stage indicates better following of pointing. Difference score was used to measure children’s understanding of giving. Larger difference scores mean better action prediction and understanding of giving in the rational giving condition. No significant relationships between following of pointing and the understanding of giving were found in children with ASD ($r = .2, p = .243$) and TD children ($r = .072, p = .676$).

Are there any correlations between giving prediction and SRS and AQ?

Partial correlation analysis with IQ as a control variable was employed to evaluate the relationship between action prediction performance in the giving stage and SRS and AQ in children with ASD. Children with ASD’s action prediction in the irrational giving condition was correlated with the communication scores in the AQ, $r = .488, p = .008$, indicating that the faster children with ASD’s predictive saccade in the irrational giving condition, the higher communication scores in the AQ indicative of worse communication in ASD children. Other sub-scores and total scores of AQ and SRS were not correlated with action prediction performance of children with ASD (all $ps > .05$).

2.3. Discussion

Experiment 1 investigated the understanding of giving in children with ASD. Results showed that children with ASD could shift their gaze to the taking hand before the giving hand arrived at the taking hand area, indicating the ability of action prediction in children with ASD might be preserved. However, it should be noted that the condition modulated TD children’s performance but not children with ASD. TD children showed faster predictive gaze in the rational giving condition relative to irrational giving condition while children with ASD’s performance did not differ in the two conditions, which suggests that children with ASD are less sensitive to the communicative meaning of giving. In addition, there was no significant difference between two groups of children in the proportional total looking time to the faces of the giver (actress A) and taker (actress B) in the two conditions, which is seemingly in contrast to prior studies that found individuals with ASD showed decreased attention to faces (Campbell et al., 2014; Rice, Moriuchi, Jones, & Klin, 2012). A possible explanation is that in experiment 1, the interaction of the two actresses was mainly focused on hand movements, and gestures contained more social information than faces. Marsh et al. (2015) explored the understanding of the rationality of actions in children with ASD, and they also found that there was no significant difference between the total fixation time to

Table 2
Demographic characteristics of participant groups in experiment 2.

	ASD ($n = 38$) M (SD)	TD ($n = 34$) M (SD)	p value
Sex, M/F	30/8	27/7	0.597
Age (years)	6.04 (0.83) Range: 5.01–8.03	6.12 (1.08) Range: 5–8	0.731
IQ	117 (17.63)	124 (11.35)	0.045 ^a
AQ-Child	Range: 90–142	Range: 100–142	
SRS	85.69 (14.78) Range: 42–113 84.69 (24.66) Range: 31–138	- - - -	

Note. AQ: Autism Spectrum Quotient-Children’s Version, SRS: Social Responsiveness Scale.

^a $p < .05$.

the faces of actors in children with ASD and TD children. This result suggests that children with ASD's attention to face is modulated by context (Kaliukhovich et al., 2020). In the social interaction scene, children with ASD pay more attention to activities and less attention to faces. Children with ASD's anticipatory looking in the irrational giving condition is associated with their communication scores in the AQ, indicating that atypical understanding of giving could be used as a precursor of impaired social communication skills in children with ASD.

3. Experiment 2

3.1. Method

3.1.1. Participants

In total, fifty-three 5–8-year-old Chinese children with ASD and forty chronological age and gender-matched TD children participated in the experiment 2. Among them, 15 ASD children and 6 TD children were excluded due to calibration failure and lack of valid trials. The final sample consisted of 38 children with ASD and 34 TD children (see Table 2). It should be noted that 18 TD children and 25 ASD children in experiment 1 and experiment 2 overlapped. The recruitment procedure of participants and standardized assessments were identical to those in experiment 1. In order to reduce the carry-over effect, we counterbalanced the order of presentation of experiment 1 and experiment 2 for those who participated in both experiments.

3.1.2. Stimuli and apparatus

The apparatus was identical to that in experiment 1. The videos showed the same action sequence as actress B pointed to the toy and then performed a taking action, and actress A grasped the toy and passed it to actress B. What was different from experiment 1 was that in experiment 2, we manipulated the hand shape of taking (rational taking or irrational taking) rather than giving, to examine children with ASD's understanding of taking (see Fig. 3). Children were presented with 12 videos with 6 rational taking trials and 6 irrational taking trials.

3.1.3. Data processing

Data processing was performed in the same way as in experiment 1. The AOIs and dependent variables were identical to those in experiment 1. We also analyzed the data in the pointing stage and taking stage separately.

3.2. Results

Can children with ASD follow the pointing gesture?

The first fixations to pointing action (hand), the object (toy), and the target of pointing (the face of actress A).

The results of one-way ANCOVA with IQ as a covariate showed that there were no significant differences in the first looking time to

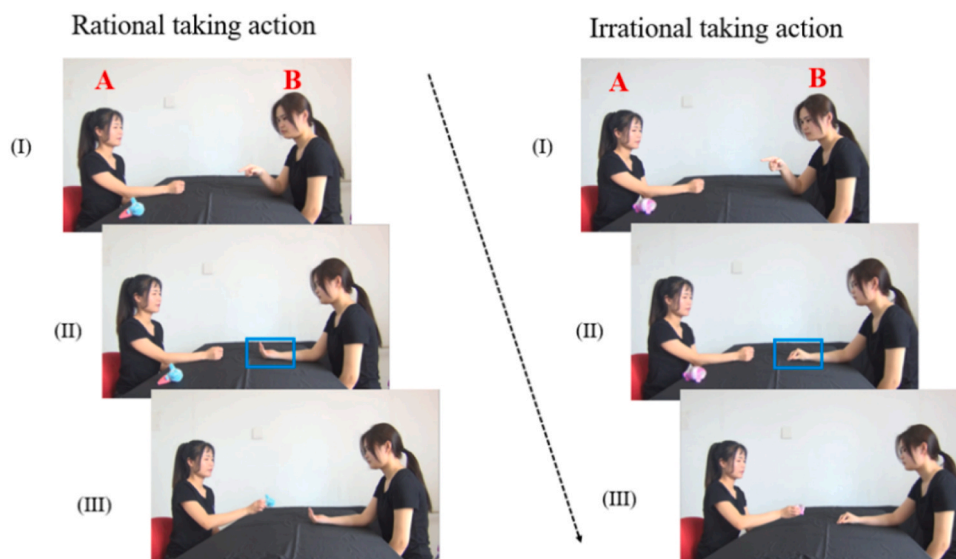


Fig. 3. Video clips of action sequences presented in experiment 2. Note. In both rational taking condition and the irrational taking condition, the actress B first pointed to the toy (I), then the actress B showed a taking gesture (II). The difference between the two conditions is that in the rational taking condition, the actress B showed a taking gesture (the action is marked by the blue rectangle in the left side) while in the irrational taking condition, the actress B showed an inverted hand shape (the action is marked by the blue rectangle in the right side). Then the actress A gave the toy to actress B (III).

pointing action ($p = .702$), the toy ($p = .086$) and the target of pointing (the face of actress A) ($p = .77$) between children with ASD and TD children.

The proportional total looking time to pointing action (hand), the object (toy), the target of pointing (the face of actress A), and the performer of pointing (the face of actress B).

The results of one-way ANCOVA with IQ as a covariate showed that children with ASD's proportional total looking time to pointing action ($p = .136$), toy ($p = .103$), the face of actress A ($p = .543$) and the face of actress B ($p = .473$) did not differ from that of TD children.

Can children with ASD understand the meaning of taking?

Difference score of TD children and children with ASD.

The mixed repeated measures ANCOVA with group (TD vs. ASD) as the between-subjects measure, condition (rational taking vs. irrational taking) as the within-subjects measure and IQ scores as the covariate was conducted to examine children's understanding of taking. The main effect of group was significant with TD children shifting their gaze to the taking action faster than children with ASD in both rational taking and irrational taking condition ($F(1, 69) = 7.655, p = .007, \eta_p^2 = .1$), as shown in Fig. 4. The main effect of condition was not significant ($F(1, 69) = .642, p = .426$), and neither was the interaction between group and condition ($F(1, 69) = .469, p = .496$).

Percentage of total looking time of two groups of children to the giving hand, the taking hand and the faces of two actresses in the rational taking and irrational taking condition.

The mixed repeated measures ANCOVA revealed a significant effect of condition ($F(1, 59) = 5.181, p = .026, \eta_p^2 = .081$) in terms of the proportional total looking time to the face of the giving person, as shown in Fig. 5. In the rational taking condition, both groups of children's proportional total fixation time to the face of the giver (actress A) was higher than that in the irrational taking condition. There were no significant differences in the proportional total looking time to the giving hand, the taking hand and the faces of the taking person between two groups of children in the two conditions (all $ps > .05$).

Is children's following of pointing correlated with understanding of taking?

Partial correlation analysis with IQ as a control variable was conducted to investigate whether children's following of pointing would influence their understanding of taking in the give-and-take interaction. We used the first fixation to the toy to measure children's following of pointing. Shorter first fixation to the toy in the pointing stage indicates better following of pointing. Difference score was used to measure children's understanding of taking. Larger difference scores mean better action prediction and understanding of taking in the rational taking condition. Results showed no significant relationships between following of pointing and the understanding of taking in both groups of children (all $ps > .05$).

Are there any correlations between taking prediction and SRS and AQ?

We assessed the relationship between the prediction of taking and the SRS and AQ scores respectively. No correlations were found between the prediction of taking and SRS total scores and subscale scores (all $ps > .05$). The relationship between the prediction of taking and AQ total scores and subscale scores were also insignificant (all $ps > .05$).

3.3. Discussion

Experiment 2 explored the understanding of taking in children with ASD. The results showed that the ability of predicting actions in children with ASD might be intact. However, children with ASD shifted their gaze to the taking hand in both conditions significantly slower than typically developing children, indicating that children with ASD show less efficiency when processing taking relative to TD children. However, it is important to note that both ASD children and TD children's prediction were not affected by conditions, which is in contrast to previous studies. For TD individuals, 12-month-old infants show sensitivity to taking, and their prediction speed in rational taking conditions was significantly faster than that in irrational taking conditions (Elsner et al., 2014). However, in our study, there were no significant main effect of condition and interaction between group and condition. As the taking task was originally

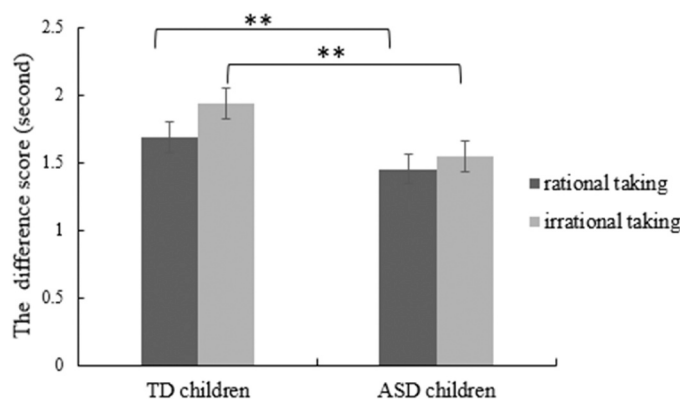


Fig. 4. Difference scores in the experiment 2. Note. The difference score was defined as the arrival time of the giving hand to the taking hand minus the arrival of gaze to the taking hand. Error bars indicate standard error. Asterisks represent statistically significant differences ($p < .01$).

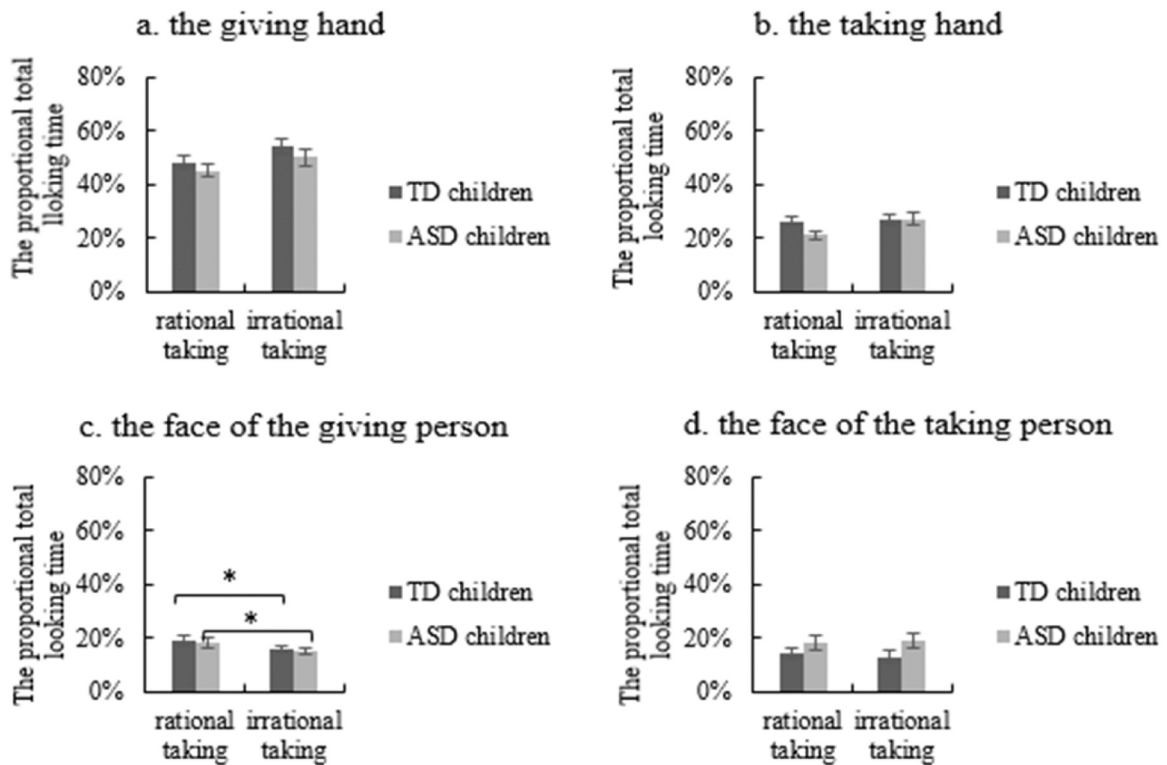


Fig. 5. Proportion of total looking time to areas of interest in the taking stage of experiment 2. Note. Error bars indicate standard error. Asterisks represent statistically significant differences ($p < .05$).

designed for TD 12-month-old infants, there may be a ceiling effect. An alternative explanation for the absence of difference in two conditions within group is that the taking gesture is not as salient as the giving gesture as it is static compared to the giving gesture. In terms of action prediction, 5–8-year-old children showed predictive saccade in both conditions, suggesting that action prediction may not be an effective measure to examine preschool children's understanding of taking. Future studies can use more sensitive and subtle measures such as EEG or pupil dilation to assess whether preschool children would be more sensitive to taking gestures compared to inverted hand shape without communicative meaning. However, although the condition failed to modulate children's performance in terms of action prediction, two groups of children had a greater proportional total fixation time to the giver's face in the rational taking condition, indicating that both groups of children expected the giver to give the toy to the taker. Therefore, they would look more at the face of the person who gave the toy and look forward to her response. These results showed that children with ASD might be able to understand the meaning of taking, although they needed to make more mental efforts than TD children.

4. General discussion

The current study investigated the understanding of giving and taking in children with ASD. We manipulated the rationality of the social actions to examine whether children with ASD generate action predictions based on the action intention or the spatiotemporal movement trajectories when they perceive goal-directed social actions, which extends prior studies as they did not discriminate this difference and mainly focused on the non-social actions. Results demonstrated that although children with ASD could predict actions, they showed less efficiency in processing give-and-take interactions. Interestingly, we found a disassociation between the understanding of giving and taking in the give-and-take interactions in children with ASD by showing impaired understanding of giving but unimpaired understanding of taking.

In terms of the understanding of giving, like TD children, children with ASD shifted their gaze to the taking hand before the giver completed the giving action, indicating that they exhibited anticipatory looking and their ability of predicting action may be preserved. This is consistent with the findings of Falck-Ytter (2010) that 5-year-old children with ASD showed predictive eye movements when observing reach-to-grasp and placement actions. These results suggest that the low-level predictive ability is unimpaired in children with ASD. However, children with ASD showed no differential predictive saccade in response to the rational giving with communicative meaning and the irrational giving without communicative meaning, indicating that they may lack sensitivity to giving. Notably, the two groups did not differ in terms of the first fixations to the socially relevant elements such as faces, gestures in the video, which reveals that the attentive bias to social information is present in children with ASD. Therefore, the altered understanding of giving in children with ASD cannot be attributed to lack of social motivation but demonstrate that children with ASD may have

impairments in processing social actions with communicative meaning. Vivanti et al. (2011) also found that it is difficult for children with ASD to use cues such as head-turn to infer other's intentions. These results reveal that although children with ASD are able to predict actions, the top-down social processing of referential cues is impaired in these children. Moreover, we identified a correlation between the prediction of irrational giving and the communication score in the AQ, indicating that attenuated understanding of giving might be a precursor of atypical social communication.

In terms of the understanding of taking, although the condition fails to identify children's understanding of taking due to potential ceiling effects, we found that the percentage of the total fixation time of two groups of children to the face of the giving person was significantly longer in the rational taking condition than that in the irrational taking condition, which suggests that both groups of children expected that the giving person would respond to the taking person. This indicates that children with ASD might be able to understand the meaning of taking, which, however, is inconsistent with Falck-Ytter et al. (2013)'s finding that when 6-year-old typically developing children observed a child performing a taking gesture, they would increase attention to the face of the person interacting with the child, but children with ASD did not show this tendency. A possible account is that the average IQ of the children with ASD in the experiment of Falck-Ytter et al. (2013) was 81 points, while the average IQ of children with ASD in our study was 117 points. The difference in the IQ of the autistic children may partly account for the contrasting results. Further work is required to investigate whether children with ASD could understand taking.

The dissociation between the understanding of giving and taking in children with ASD reflects that the essential property of giving and taking might be different. It has been suggested that gestures that serve to request objects or actions are imperative gestures while gestures that share interest in an object or convey information about it are declarative gestures (Bates et al., 1975; Carpenter et al., 2002). There is evidence that the understanding of imperative gestures is intact in children with ASD whereas the understanding of declarative gestures is absent in children with ASD (Baron-Cohen, 1989; Carpenter et al., 2002). In the current study, giving that means sharing objects is a declarative gesture and taking that refers to 'give me' or 'I want that' is an imperative gesture. Our finding that the understanding of giving is impaired while the comprehension of taking is intact corroborates with previous studies (Baron-Cohen, 1989; Carpenter et al., 2002). As declarative gestures require shared attention or intent while imperative gestures do not necessarily require shared intent, the failure to understand declarative gestures in children with ASD might be associated with a deficit in shared intention. In addition, Tatone et al. (2015) found that in human brain, taking is represented as an object-oriented non-social action while giving is represented as a patient-directed social action. Compared with taking, the perception of giving elicited stronger lower alpha suppression that reflects the perceived level interactivity of observed actions (Yin et al., 2020). Taken together, these findings suggest that children with ASD have difficulty processing social actions involving shared intention or attention, which might be related to impaired theory of mind in this population (Baron-Cohen, 1989) as accumulating evidence shows the reduced functional connectivity in the mentalizing networks (i.e., the frontal and posterior temporal regions and temporoparietal junction) when interpreting others' action intention (Cole et al., 2019; Kana et al., 2014; Libero et al., 2014).

There are several limitations to our study. First, the IQ scores of TD children were significantly higher than that of children with ASD in our study. However, after we controlled the influence of IQ, group differences remain unchanged in terms of action prediction. In order to obtain more solid and stable evidence, ASD children and IQ-matched TD children should be tested in the future research. Second, children with ASD are older in our study and have much experience with give-and-take interactions in daily life, which may impact their performance. And little is known about whether children with ASD show impairments in the understanding of giving and taking from early childhood. Future work is required to investigate the understanding of giving and taking in infants at elevated risk for autism (i.e., high risk infants whose older siblings are diagnosed as ASD), which has great implications for early diagnosis and targeted interventions. Third, we did not calculate the application reliability of the procedure and future work is needed to replicate the eye tracking procedure in the current study to ensure the reliability. Moreover, despite being capable of measuring the online action understanding, the eye tracking technique is unable to reveal the neural basis of action understanding in children with ASD. Future research would benefit from combining the eye tracking measures with brain-based measures (i.e., Event-related potential, fMRI) to unveil the mechanisms underlying action understanding in children with ASD. Fourth, due to the small sample size and the overlap of samples in the two experiments, it should be cautious to interpret and generalize the current findings although the results in the two experiments remained unchanged after the exclusion of the same samples. Future work is required to increase the sample size and adopt between-groups design to examine the understanding of giving and taking in children with ASD.

5. Conclusion

The current findings suggest that the low-level action prediction of children with ASD is preserved while they show reduced sensitivity to the giving gesture and less efficiency when processing the give-and-take interactions. More importantly, the current study reveals that the understanding of goal-directed social actions is not entirely impaired in children with ASD, and the altered understanding of giving might be a precursor of atypical social communication. Our study provides new insights for understanding the mechanism of impaired social communication, especially prosocial behavior in children with ASD.

Ethics

This study was approved by the Institute of Psychology, Chinese Academy of Sciences Child Subjects Committee and conformed to the ethical standards set by the Declaration of Helsinki.

CRediT authorship contribution statement

Hou Wenwen: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft. **Li Jing:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ridd.2023.104642](https://doi.org/10.1016/j.ridd.2023.104642).

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